

$$\textcircled{1} \quad T_n^\dagger \hat{n} T_n = \exp\left(\frac{i\hat{p}x}{\hbar}\right) \hat{n} \exp\left(\frac{-i\hat{p}x}{\hbar}\right)$$

$$\exp(A) B \exp(-A)$$

$$= B + [A, B] + [A, [A, B]]$$

$$\begin{aligned} \left[\frac{i\hat{p}x}{\hbar}, \hat{n} \right] &= \frac{i\hat{p}x}{\hbar} \cdot \hat{n} - \hat{n} \frac{i\hat{p}x}{\hbar} \\ &= \frac{i\cancel{\hbar} \cancel{\partial x}}{\cancel{\hbar} \cancel{\partial x}} \hat{n} - \frac{\hat{n} i\cancel{\hbar} \cancel{\partial x}}{\cancel{\hbar} \cancel{\partial x}} = 0 \\ &= \underline{\underline{\hat{n}}} \end{aligned}$$

$$\begin{aligned} [T_n, \hat{p}] &= \exp\left(\frac{-i\hat{p}\hat{n}}{\hbar}\right) \exp\left(\frac{i\hat{p}\hat{n}}{\hbar}\right) \\ &\quad - \exp\left(\frac{i\hat{p}\hat{n}}{\hbar}\right) \exp\left(\frac{-i\hat{p}\hat{n}}{\hbar}\right) \\ &= 1 - 1 = \underline{\underline{0}} \end{aligned}$$

$$\textcircled{2} \quad \langle \hat{n} | \hat{p} | \psi \rangle = \frac{\hbar}{i} \frac{d}{dx} \langle x | \psi \rangle$$

$$\langle x | \hat{p}^n | \psi \rangle = \left(\frac{\hbar}{i} \frac{d}{dx} \right)^n \psi(x)$$

$$\langle x | \left(\frac{\hbar}{i} \frac{d}{dx} \right)^n | \psi \rangle = \left(\frac{\hbar}{i} \frac{d}{dx} \right)^n \underbrace{\langle x | \psi \rangle}_{\psi(x)}$$